Comparative Biochemistry and Physiology, Part C 290 (2025) 110144

Contents lists available at ScienceDirect

Comparative Biochemistry and Physiology, Part C

journal homepage: www.elsevier.com/locate/cbpc

Individual and combined effects of microplastics and diphenyl phthalate as plastic additives on male goldfish: A biochemical and physiological investigation

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ARTICLE INFO

Edited by Martin Grosell

Keywords:
Microplastics
Plastic additives
Synergistic effects
Goldfish
Biochemical biomarkers
Endocrine disruption

ABSTRACT

The development of the plastics industry worldwide has led to an increase in the rate of plastic waste and chemical additives such as microplastics (MPs) and diphenyl phthalate (DPP) in the environment. The penetration of these pollutants into aquatic ecosystems has also raised concerns about their toxic effects, individually and in combination. The present study investigated the individual and combined toxicity of MPs and DPP on the health of male goldfish. A 28-day exposure experiment was conducted using different concentrations of DPP (2.5, 5.0, 7.5 μL L^{-1}) and MPs (20, 40 mg L^{-1}), both individually and in combination. Biochemical markers, enzyme activities, and hormone levels were evaluated to ascertain the effects on metabolic, renal, and reproductive health. The findings revealed that concurrent exposure to DPP and MPs markedly elevated plasma glucose, creatinine, triglycerides, and cholesterol levels, accompanied by notable reductions in high-density lipoprotein and low-density lipoprotein. Moreover, combined exposures resulted in liver damage, as evidenced by elevated serum glutamic-oxaloacetic transaminase, serum glutamic-pyruvic transaminase, alkaline phosphatase, lactate dehydrogenase, and gamma-glutamyl transferase activities and disruptions in protein synthesis and immune response, with notable decreases in total protein, albumin, and globulin. Testosterone levels decreased, while estradiol levels increased, indicating endocrine disruption and potential reproductive impairment. These findings indicated the adverse synergistic effects of MPs and DPP on the physiology of goldfish. Therefore, further research must be conducted to increase our knowledge of their ecotoxicological risks.

1. Introduction

The development of plastic and polymer industries worldwide has increased various environmental pollutants (Ali et al., 2024; Arif et al., 2024; Banaee et al., 2023a, 2024a; Burgos-Aceves et al., 2024; C.E. Choi et al., 2022), showing that leachates from plastic manufacturing facilities might contain 240 additives, mainly plasticisers, antioxidants, and flame retardants (Dobrzycka-Krahel et al., 2024; Gholamhosseini et al., 2024; Qiu et al., 2022; Zeidi et al., 2023; Zhang et al., 2019).

In this context, diphenyl phthalate (DPP) is a chemical compound commonly used as a plasticizer to enhance the flexibility and durability of plastic products, such as vinyl flooring, toys, and food packaging (Johnsi and Suthanthiraraj, 2018). However, some countries have restricted its use due to concerns about potential health and environmental effects (Teil et al., 2014). DPP has been detected in surface water, including rivers, lakes, and streams (Luo et al., 2021). The presence of DPP in surface water can be attributed to various sources, such as wastewater discharge, industrial runoff, and atmospheric deposition. Once released into the environment, DPP can undergo degradation processes, although it can persist for some time. Its ability to partition between water and sediment can vary depending on environmental conditions (Wu et al., 2019; Zhou et al., 2017). The presence of DPP in

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https://doi.org/10.1016/j.cbpc.2025.110144
Received 2 December 2024; Received in revised form 19 January 2025; Accepted 6 February 2025
Available online 7 February 2025

surface water raises concerns about its potential impacts on aquatic organisms and ecosystems. (Lu et al., 2021) found estrogenic endocrine-disrupting chemicals, including phthalates, BPA, NP, and MPs, in fish from northern Taiwan's estuaries. Studies have indicated that DPP can adversely affect aquatic organisms, including fish, invertebrates, and algae. (Jebara et al., 2021) found that non-phthalate plasticisers were more prevalent than phthalate esters, with di(2-ethylhexyl) phthalate and di(2-ethylhexyl) terephthalate being the most abundant in both water and sediment. (Qiu et al., 2022) found that plasticisers, antioxidants, and flame retardants could inhibit bioluminescence and induce abnormal behaviour in marine medaka larvae. DPP is classified as an endocrine disruptor that can interfere with the normal functioning of the endocrine system, which regulates hormones in the animals' body. Studies have shown that DPP has the potential to disrupt hormonal balance and affect reproductive and developmental processes in aquatic animals. For instance, (Shu et al., 2024) found that 2-ethylhexyl diphenyl phosphate disrupts thyroid hormone function in zebrafish (*Danio rerio*). The specific effects of DPP on fish would depend on various factors such as the concentration, duration of exposure, and the species of fish involved. However, studies have indicated potential adverse effects of phthalates, including DPP, on fish. Indeed, DPP can disrupt the endocrine system and interfere with the reproductive processes of fish, leading to impairments in fertility, gonadal development, and hormone production. (Tseng et al., 2021, 2023) showed that waterborne bisphenol A (BPA) and diethyl phthalate (DEP) disrupt the reproductive capacity of zebrafish, impacting the development and swimming behaviour of their offspring. BPA exposure elevated ovarian 17β-estradiol levels and hepatic vitellogenin expression, while DEP decreased these levels. BPA increased the expression of esr2a, whereas DEP reduced esr1 and esr2b levels. Exposure to DPP may also impact fish behaviour and physiology. Studies have shown that phthalates can alter fish swimming behaviour, feeding patterns, and growth rates (Tseng et al., 2023; Hsu et al., 2025; Huang et al., 2023). Zhou et al. (Zhou et al., 2017) showed that DEP, BPA, and estradiol, as endocrine-disrupting chemicals, could disrupt ceratohyal cartilage development in embryos when the maternal generation was exposed to specific doses. They found that BPA and DEP may disrupt estrogen signaling and influence neural crest cell migratory genes. Moreover, Chen et al. (Chen et al., 2021) found that parental co-exposure to DBP and DiBP in zebrafish increased hatchability and heart rate and led to more malformations and mortality in offspring. Ogunwole et al. (Ogunwole et al., 2024) showed that ibuprofen and dibutyl phthalate altered gene expression in *Clarias gariepinus*, upregulating growth genes and downregulating forkhead Box Protein L2, and cytochrome P450, indicating potential sex reversal. Ibuprofen had the strongest binding affinity to docked genes. Furthermore, they may affect biochemical processes in fish, such as metabolic activities and enzyme functions. Huang et al. (Huang et al., 2023) found that BPA and BPAF could alter the FoxO and MAPK pathways, while BPS influenced energy metabolism. Moreover, DPP exposure during critical stages of fish development, such as embryonic or larval, can adversely affect growth, development, and survival rates. It may lead to developmental abnormalities and impairments in organ systems. Phthalates tend to accumulate in fish tissues through bioaccumulation and biomagnification processes. This accumulation can have long-term effects on fish health and potentially impact higher levels of the food chain if fish contaminated with DPP are consumed by other organisms. Although DPP is not typically found in the form of MPs, it can be released through the breakdown of larger plastic items or plastic polymers. DPP, as a plasticizer in certain plastic products, can contribute to the overall presence of MPs in the environment (Banaee et al., 2024a).

MPs in surface water have become a growing concern worldwide. Studies have shown that MPs can enter surface water bodies through various sources, including the breakdown of larger plastic debris, microbeads from personal care products, and fibres from synthetic textiles (Banaee et al., 2024a, 2023b; Burgos-Aceves et al., 2024; Bobori et al., 2022; Capó et al., 2022; Chen et al., 2020). When plastic products degrade or break down, they can release MPs into the environment, including bodies of water. MPs can accumulate in aquatic environments and have detrimental effects on ecosystems. Thus, various organisms, including fish, shellfish, and plankton, may ingest MPs, causing physical harm, reduced feeding capability, and potential transfer of toxins through the food chain. Moreover, MPs can affect the quality of surface water. They can also alter water chemistry, adsorb and transport harmful pollutants, and act as vectors for transporting microorganisms or invasive species (C.E. Choi et al., 2022; H. Choi et al., 2022; Dent et al., 2023; Ding et al., 2018; Gholamhosseini et al., 2023).

This study hypothesized that microplastics (MPs) and diphenyl phthalate (DPP), a plastic additive, could cause toxicity in fish on their own. Additionally, MPs could act as a vector to enhance the toxic effects of DPP in fish. Furthermore, DPP, as an endocrine-disrupting compound, could cause reproductive and biochemical abnormalities in male fish. Therefore, the novelty of our study lies in the simultaneous exposure to MPs and DPP, which was predicted to cause synergistic toxicity, leading to more significant biochemical, metabolic, and endocrine disruptions compared to individual exposures. This study aimed to investigate the impact of MPs and DPP on goldfish health. The results may help increase our understanding of the effects of these pollutants on goldfish health and the risks associated with exposure to MPs and DPP, both individually and in combination.

2. Materials and methods

2.1. Chemicals

Diphenyl phthalate (DPP), with the chemical formula $C_6H_4(CO_2C_4H_9)_2$ and a purity of 99 %, was obtained from Chimi Tejart Maham Co., (Tehran) Iran. A powder form of microplastics (MPs), including high-density polyethylene (HDPE-MPs) with an average size smaller than 0.02 mm (15–25 μm) and a density of 0.975 g cm^{-3} was purchased from Ishraq Iran Trading Company based in Shiraz, Iran. Biochemical reagents and oxidative biomarker kits were acquired from Biorex-Fras (Shiraz, Iran) and Kiazist Company (Hamadan, Iran).

2.2. Fish

Adult male goldfish with 35 ± 5 g mean body weight were bought from a local farm, transferred into the laboratory, distributed into 24 aquariums, and maintained in the laboratory conditions for 14 days. The temperature was maintained at 26 ± 2 °C, pH level at 7.4 ± 0.2, and dissolved oxygen at 7.0 ± 0.5 mg L^{-1}. The goldfish were given ornamental fish feed twice a day during the adaptation period. The specific brand used was Beyza Feed Mill from Shiraz, Iran. The feed contained 45–55 % protein, 10–11 % lipid, 20–30 % carbohydrate, and 1.5–2 % fibre. The Ethics Committee of the Behbahan Khatam Alanbia University of Technology in Iran approved the experimental procedure and methodology of this study (Approval ID: 6–1-112,667/13–03-02).

2.3. Experimental design

One hundred and forty-four male goldfish were subjected to different concentrations of Diphenyl phthalate (DPP) and microplastics (MPs). The control group (Group I) consisted of fish that were not exposed to DPP or MPs. Group II was exposed to DPP at a concentration of 2.5 μL L^{-1}, while Group III was exposed to DPP at a concentration of 5 μL L^{-1}. Group IV experienced DPP exposure at a concentration of 7.5 μL L^{-1}. Fish in Group V and Group VI were exposed to MPs at concentrations of 20 mg L^{-1} and 40 mg L^{-1}, respectively. Group VII to Group XII were exposed to combinations of DPP and MPs at different concentrations. The specific combinations and concentrations varied across these groups (2.5 μL L^{-1} mixed with 20 or 40 mg L^{-1} MPs; 5 μL L^{-1} mixed with 20 or 40 mg L^{-1} MPs; 7.5 μL L^{-1} mixed with 20 or 40 mg L^{-1} MPs).

After 28 days of exposure to contaminants, fish were anaesthetized using 150 mg L^{-1} of clove powder. Blood samples were collected from the caudal vein using heparinized sterile syringes and stored in 2 mL Eppendorf tubes. The samples were then centrifuged (4 °C, 15 min, 6000 g), and the resulting supernatant was used for blood biochemical assessments.

2.4. Biochemical parameters

The study measured several biochemical parameters, enzyme activities, and sexual hormones in the goldfish plasma exposed to DPP and MPs. The parameters analyzed in the study encompassed glucose, total protein, albumin, cholesterol, triglyceride, high-density lipoprotein, low-density lipoprotein, and creatinine. Furthermore, the study also investigated enzyme activities, including serum glutamic-oxaloacetic transaminase, serum glutamic-pyruvic transaminase, gamma-glutamyl transpeptidase, alkaline phosphatase, and lactate dehydrogenase. Moreover, the study also assessed the levels of testosterone and estradiol hormones in the plasma. All the measurements were carried out using diagnostic biochemical reagents from Biorex-Fars Co. (Shiraz, Iran) and were analyzed with a UV/Vis spectrophotometry instrument (Unico, 2100). The concentrations of testosterone and estradiol hormones were evaluated using the Mono Kit (Produced by Yasin Tab Company (Tehran, Iran) under MonoBind Inc. license) and the ELISA analyzer (Dana-3200 model from Garni Medical Engineering Co., Tehran, Iran).

2.5. Assessment of synergism and antagonism

The possible interaction between microplastics (MPs: 20 and 40 mg L^{-1}) and diphenyl phthalate (DPP: 2.5, 5.0, and 7.5 μL L^{-1}) effects were estimated according to the following mathematical models (Aliko et al., 2015; Banaee et al., 2020, 2023c, 2023d; Sula et al., 2020). The synergism rate only occurs when the expected effect is greater than the observed effect.

1. Predicted effect of the endpoints of fish exposed to individual MPs and DPP

$$Predicted\ effect = \frac{MPs}{Control} \times \frac{DPP}{Control}$$

2. Observed effect of the endpoints of fish exposed to MPs and DPP combination

$$Observed\ effect = \frac{The\ combination\ of\ MPs\ and\ DPP}{Control}$$

3. The synergistic effect

$$Synergy\ ratio = \frac{Predicted\ effect}{Observed\ effect}$$

2.6. Statistical analysis

To validate the normality of the data, the Kolmogorov-Smirnov test was performed. Next, the impact of two independent variables, including gender and the concentrations of Diphenyl phthalate (DPP) and microplastics (MPs), on blood biochemical parameters and sexual hormones was assessed through Two-way analysis of variance (Two-way ANOVA). Tukey test was employed to compare means in order to confirm statistical significance with a confidence level of 99 % ($p = 0.01$). The results were presented as mean ± standard deviation, with each value based on a sample size of 9. Mean values within groups that do not share the same letter were considered significantly different ($p < 0.01$). All statistical analyses were conducted using GraphPad Prism 8.4.2.

3. Results

Table 1 shows results relative to the evaluation of synergism of MPs and DPP concerning the biochemical parameters, including glucose, creatinine, triglycerides, cholesterol, high-density lipoprotein (HDL), and low-density lipoprotein (LDL). Exposure to 5 μL L^{-1} DPP combined with 40 mg L^{-1} MPs and 7.5 μL L^{-1} DPP mixed with 20 or 40 mg L^{-1} MPs increased glucose in the goldfish plasma. However, DPP or MPs alone could not change the glucose levels. Treatment of fish with 20 mg L^{-1} MPs increased creatinine levels in the goldfish plasma. Although no significant change was detected in the creatinine levels in the plasma of goldfish exposed to 2.5 μL L^{-1} DPP combined with 40 mg L^{-1} MPs, co-exposure to DPP and MPs increased the creatinine levels. Results showed that exposure to DPP did not significantly affect triglyceride levels in the goldfish plasma. Although 20 mg L^{-1} MPs could not change triglyceride contents, exposure to 40 mg L^{-1} MPs increased triglyceride concentrations in the goldfish plasma. Triglyceride levels in the plasma of goldfish exposed to DPP mixed with MPs were significantly higher than in the control group. Although no significant difference was observed in cholesterol levels in fish exposed to 2.5 and 5 μL L^{-1} DPP, exposure to 7.5 μL L^{-1} increased cholesterol contents. Also, cholesterol levels were significantly increased in the plasma of goldfish exposed to MPs alone and combined with DPP. Although HDL levels were significantly increased in the treated fish with 2.5 μL L^{-1} DPP compared with the control group, exposure to 7.5 μL L^{-1} DPP decreased HDL contents, and contact with MPs decreased HDL levels. Exposure to 5 μL L^{-1} DPP mixed with 40 mg L^{-1} MPs or 7.5 μL L^{-1} combined with 20 or 40 mg L^{-1} MPs could decrease HDL levels in fish plasma. Although exposure to MPs and DDP alone could not change LDL levels, co-exposure to MPs and DDP (except fish exposed to 2.5 μL L^{-1} DPP mixed with 20 mg L^{-1} MPs) decreased LDL concentrations in the plasma of goldfish (Fig. 1).

For cholesterol levels, a synergistic effect was observed at higher concentrations of MPs and DPP (7.5 μL L^{-1} DPP and 40 mg L^{-1} MPs), while other combinations showed additive effects. Similarly, triglycerides exhibited primarily additive effects in all treatments. High-density lipoprotein (HDL) levels were reduced in the combined treatments, showing additive effects at lower concentrations and a suppressive effect at the highest concentrations (7.5 μL L^{-1} DPP and 40 mg L^{-1} MPs). Low-density lipoprotein (LDL) levels showed predominantly additive effects, with a minor synergistic effect observed at the highest combination (7.5 μL L^{-1} DPP and 40 mg L^{-1} MPs). For glucose, the combined exposure led to primarily additive effects, indicating a mild increase in plasma glucose levels. In contrast, creatinine levels demonstrated a synergistic effect at the lowest combination (2.5 μL L^{-1} DPP and 20 mg L^{-1} MPs), while other combinations exhibited additive or suppressive effects.

Table 2 shows the results of the evaluation of synergism of MPs and DPP concerning the biochemical parameters, including total protein, albumin, globulin, and total immunoglobulins of male goldfish. Results showed that total protein levels in the goldfish plasma exposed to 7.5 μL L^{-1} DPP were significantly lower than in the control group. The presence of a mixture of 2.5 μL L^{-1} DPP and MPs, as well as 5 μL L^{-1} DPP combined with 40 mg L^{-1} MPs in water, decreased total protein levels. Exposure to MPs in combination with different concentrations of DPP increased albumin levels in the plasma of goldfish. Results showed no significant changes in globulin levels in the treated fish with 2.5 and 5.0 μL L^{-1} DPP, while 7.5 μL L^{-1} DPP could decrease globulin levels. Moreover, exposure to MPs alone and combined with DPP reduced globulin. No significant difference was observed in immunoglobulin levels in the plasma of fish exposed to MPs alone and 2.5 and 5.0 μL L^{-1} DPP compared to the control group. However, immunoglobulin levels in fish plasma exposed to 7.5 μL L^{-1} DPP were significantly lower than in the control group. Furthermore, exposure to 2.5 μL L^{-1} DPP combined with MPs and 5 μL L^{-1} DPP combined with 20 mg L^{-1} MPs resulted in

Table 1
Evaluation of synergism of microplastics (MPs) and diphenyl phthalate (DPP) concerning the biochemical parameters (e.g., glucose, creatinine, triglycerides, cholesterol, high-density lipoprotein (HDL), and low-density lipoprotein (LDL)) of male goldfish.

Parameters	Treatments	Predicated effect	Observed effect	Synergy ratio	Combined effect
Cholesterol	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.66	1.61	1.03	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.75	1.68	1.04	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.64	1.65	1.00	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.74	1.55	1.12	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.63	1.60	1.64	S
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.78	1.58	1.76	S
Triglycerides	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.22	1.30	0.94	A
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.21	1.38	0.87	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.25	1.48	0.84	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.24	1.44	0.86	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.24	1.64	0.75	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.22	1.65	0.74	A
High-density lipoprotein (HDL)	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.89	0.81	1.10	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.88	0.73	1.20	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.79	0.77	1.02	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.78	0.68	1.15	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.42	0.70	0.60	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.41	0.69	0.60	A
Low-density lipoprotein (LDL)	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.37	1.73	0.79	A
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.67	2.00	0.83	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.41	1.90	0.74	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.72	2.07	0.83	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.91	2.07	0.93	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.33	2.07	1.13	D
Glucose	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.12	1.42	0.79	A
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.27	1.38	0.92	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.11	1.40	0.79	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.26	1.75	0.72	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.49	1.86	0.80	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.69	1.84	0.92	A
Creatinine	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	3.36	1.84	1.82	S
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.27	3.31	0.68	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	3.09	2.61	1.18	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.08	2.67	0.78	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	4.28	2.92	1.47	D
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.89	2.80	1.03	D

Combined effects: CF; additive effects ($1 \leq CF \leq 1.5$): D; synergistic effect: ($CF \geq 1.5$) S; suppressive effect ($CF \leq 0.99$): A.

decreased immunoglobulin levels (Fig. 2).

For total protein, the combination of DPP and MPs generally exhibited additive effects (CF ranging from 0.99 to 1.11), indicating that the pollutants did not significantly enhance or suppress protein levels when exposed together. Similar patterns were observed for albumin, with combined exposure leading to additive effects (CF values between 0.87 and 0.94), suggesting minimal alteration in albumin levels due to combined pollutant exposure. Globulins showed more variability, with lower concentrations of DPP and MPs (e.g., 2.5 μL L^{-1} DPP and 20 mg L^{-1} MPs) producing synergistic effects ($CF \geq 1.5$), indicating a pronounced impact on globulin levels when both pollutants were combined. However, at higher concentrations (e.g., 7.5 μL L^{-1} DPP and 40 mg L^{-1} MPs), the effects became additive ($CF \leq 1.5$), suggesting a reduced impact at higher doses.

Total immunoglobulins similarly exhibited mostly additive effects (CF between 0.85 and 1.10), with no significant enhancement or suppression observed in the combined exposure treatments.

Table 3 shows the evaluation of synergism of MPs and DPP concerning the plasma enzyme activities including SGOT, SGPT, ALP, LDH, and GGT in male goldfish.

Although no significant difference was detected in the SGOT activities in the plasma of fish exposed to MPs and DDP, separately, compared to the control group, co-exposure to 7.5 μL L^{-1} DDP and MPs, as well as 5 μL L^{-1} DDP and 40 mg L^{-1} MPs increased SGOT activities. The results showed that SGPT activities in the plasma of fish co-exposed to 7.5 μL L^{-1} DDP and different concentrations of MPs were significantly higher than in the control group. A significant increase in ALP activities was observed in the plasma of fish co-exposed to MPs and DDP. However, exposure to MPs and DDP alone could not change ALP activities. Although no significant change was found in LDH activities in the plasma of fish exposed to 2.5 μL L^{-1} DDP when compared with the control group, LDH activities were significantly increased in the other treatments. The highest activities of LDH were observed in the plasma of fish exposed to 7.5 μL L^{-1} DDP combined with MPs. GGT activities were significantly increased following fish exposure to 7.5 μL L^{-1} alone and in combination with MPs. However, GGT activities were not changed in the other treatments compared with the control group (Fig. 3).

The study assessed the synergistic effects of MPs and DPP on SGOT, SGPT, ALP, LDH, and GGT activities in the plasma of male goldfish. SGOT and SGPT showed mainly additive effects, with a few suppressive results. ALP exhibited a mix of additive and suppressive effects. LDH demonstrated synergistic effects across all combinations, indicating enhanced enzyme activity. GGT showed predominantly additive effects, with a few combinations slightly below the additive threshold (Table 3).

Results showed that estradiol levels in the plasma of fish exposed to 2.5 μL L^{-1} DDP were not significantly different from those in the control group, whereas estradiol levels were significantly increased in the other treatments. The highest estradiol concentrations were found in the plasma of fish exposed to 5 and 7.5 μL L^{-1} DDP combined with 40 mg L^{-1} MPs. Although no significant changes were observed in the testosterone levels in the plasma of fish exposed to 40 mg L^{-1} MPs, exposure to 40 mg L^{-1} MPs significantly decreased their levels. Moreover, testosterone levels were significantly decreased in the plasma of fish exposed to different concentrations of DPH. The lowest testosterone concentrations were observed in fish exposed to 7.5 μL L^{-1} DDP mixed with 40 mg L^{-1} MPs (Fig. 4).

For testosterone levels, the combined exposure of 2.5 μL L^{-1} DPP and 20 mg L^{-1} MPs showed a synergistic effect ($CF = 1.66$), while the

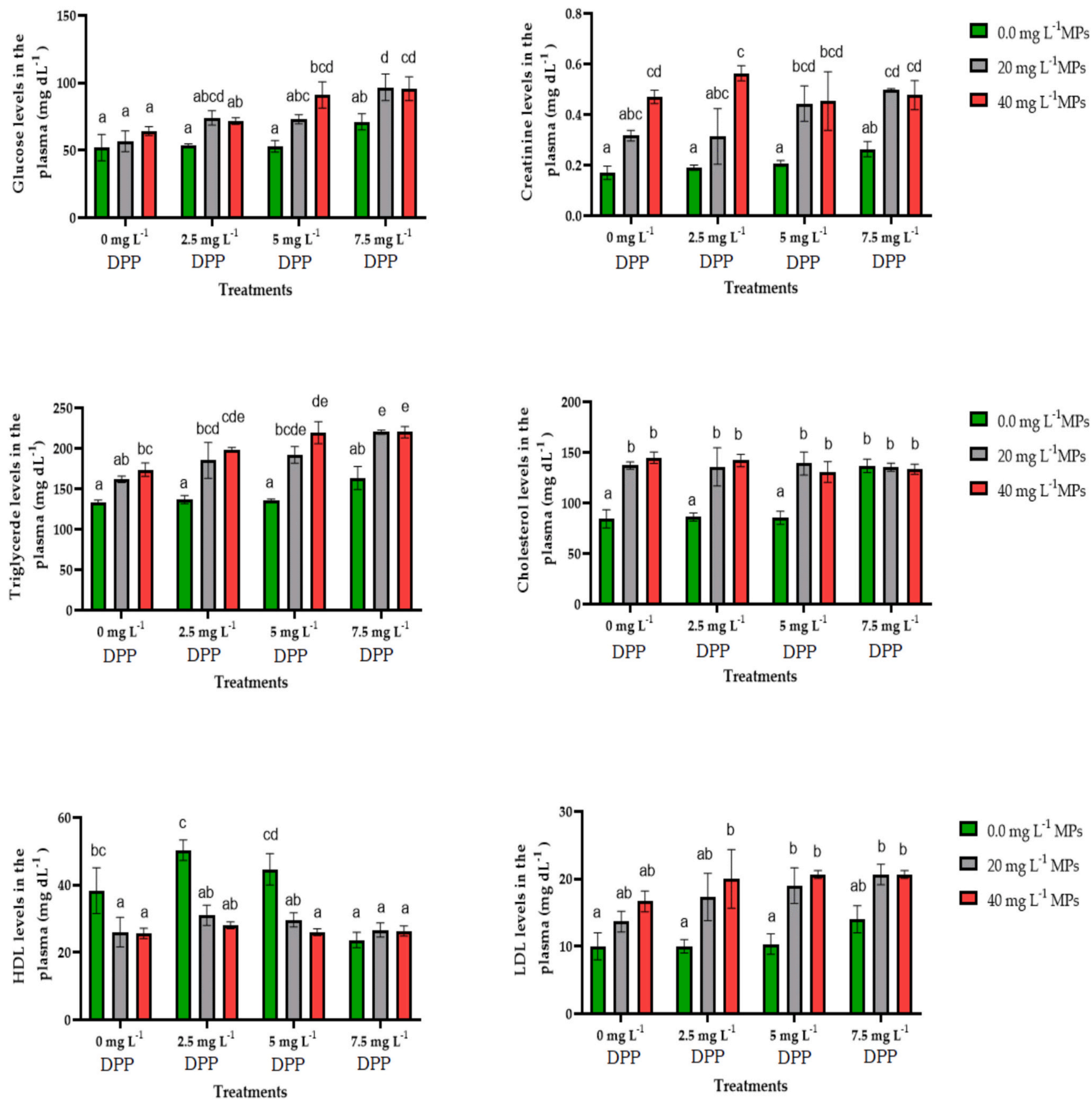

Fig. 1. Plasma biochemical parameter levels (e.g., glucose, creatinine, triglycerides, cholesterol, high-density lipoprotein (HDL), and low-density lipoprotein (LDL)) of male goldfish exposed to varying concentrations of microplastics (MPs: 0.0, 20 and 40 mg L^{-1}) and diphenyl phthalate (DPP: 0.0, 2.5, 5.0, and 7.5 μL L^{-1}), both individually and in combination, over 28 days. Different letters above the columns indicate statistically significant differences between experimental groups based on Tukey's test ($p < 0.01$), while the same letters indicate no significant difference.

combination of 2.5 μL L^{-1} DPP and 40 mg L^{-1} MPs demonstrated an additive effect (CF = 1.08). Similar results were observed for other concentrations, with the 5 μL L^{-1} DPP and 20 mg L^{-1} MPs combination showing a synergistic effect (CF = 1.62) and the 5 μL L^{-1} DPP and 40 mg L^{-1} MPs combination exhibiting an additive effect (CF = 1.03). The 7.5 μL L^{-1} DPP and 20 mg L^{-1} MPs combination also showed a synergistic effect (CF = 1.58), while the combination of 7.5 μL L^{-1} DPP and 40 mg L^{-1} MPs demonstrated a suppressive effect (CF = 0.93). For estradiol levels, combinations of DPP and MPs at various concentrations resulted in either additive or suppressive effects. Notably, the combination of 2.5 μL L^{-1} DPP and 20 mg L^{-1} MPs showed a suppressive effect (CF = 0.44), while the 2.5 μL L^{-1} DPP and 40 mg L^{-1} MPs combination exhibited a suppressive effect (CF = 0.93). Other combinations showed similar results, with several exhibiting additive effects and others showing a dose-dependent pattern (Table 4).

4. Discussion

Several recent studies demonstrated that the growth in the manufacture and utilisation of plastic products over recent years has resulted in a concomitant increase in the release of plastic waste into the environment. The degradation of plastic products by biological, chemical, and physical agents produces MPs (Aliko et al., 2022; Alimba et al., 2021; Debnath et al., 2024; Hodkovicova et al., 2022; Shirmohammadi et al., 2024). Furthermore, additives used to synthesize plastic polymers may inadvertently enter the environment. Therefore, these materials

Table 2
Evaluation of synergism of microplastics (MPs) and diphenyl phthalate (DPP) concerning the biochemical parameters (e.g., total protein, albumin, globulin, and total immunoglubolins) of male goldfish.

Parameters	Treatments	Predicated effect	Observed effect	Synergy ratio	Combined effect
Total protein	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.88	0.81	1.09	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.86	0.80	1.07	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.90	0.81	1.11	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.87	0.88	0.99	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.74	0.86	0.86	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.72	0.88	0.82	A
Albumin	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.13	1.22	0.93	A
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.12	1.24	0.90	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.17	1.24	0.94	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.15	1.32	0.87	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.29	1.30	0.99	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.27	1.32	0.96	A
Globulins	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.58	0.35	1.69	S
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.57	0.31	1.83	S
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.59	0.33	1.81	S
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.58	0.38	1.51	S
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.31	0.36	0.85	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.30	0.38	0.78	A
Total immunoglobulins	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.90	0.81	1.10	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.86	0.81	1.06	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.90	0.81	1.10	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.86	0.90	0.96	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.76	0.86	0.88	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.73	0.86	0.85	A

Combined effects: CF; additive effects (1 ≤ CF ≤ 1.5): D; synergistic effect: (CF ≥ 1.5) S; suppressive effect (CF ≤ 0.99): A.

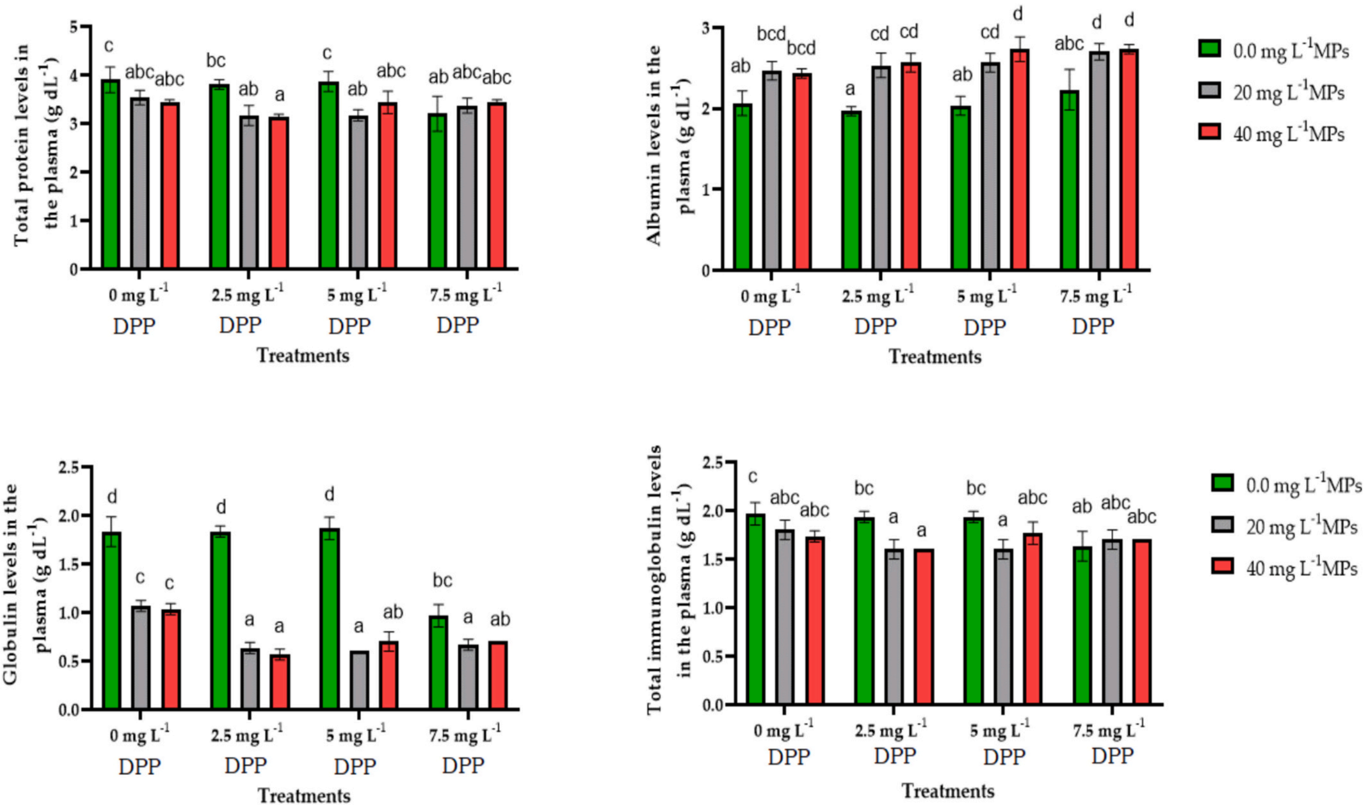

Fig. 2. Plasma biochemical parameter levels (e.g., total protein, albumin, globulin, and total immunoglubolins) of male goldfish exposed to varying concentrations of microplastics (MPs: 0.0, 20 and 40 mg L^{-1}) and diphenyl phthalate (DPP: 0.0, 2.5, 5.0, and 7.5 μL L^{-1}), both individually and in combination, over 28 days. Different letters above the columns indicate statistically significant differences between experimental groups based on Tukey's test ($p < 0.01$). Columns with the same letters indicate no significant difference.

can be a potential threat to aquatic ecosystems (Hermabessiere et al., 2017; Liu et al., 2020; Multisanti et al., 2022; Zicarelli et al., 2023). According to the study, when exposed to a combination of 5 μL L^{-1} DPP and 40 mg L^{-1} MPs or 7.5 μL L^{-1} DPP mixed with either 20 mg L^{-1} or 40 mg L^{-1} MPs, the goldfish exhibited increased glucose in their plasma. This result suggested that the concurrent presence of DPP and MPs had a synergistic effect on glucose levels. Results showed that DPP and MPs might significantly affect processes involved in glucose metabolism,

Table 3
Evaluation of synergism of microplastics (MPs) and diphenyl phthalate (DPP) concerning the plasma enzyme activities (e.g., serum glutamic-oxaloacetic transaminase (SGOT), serum glutamic-pyruvic transaminase (SGPT), alkaline phosphatase (ALP), lactate dehydrogenase (LDH), and gamma-glutamyl transferase (GGT)) of male goldfish.

Parameters	Treatments	Predicated effect	Observed effect	Synergy ratio	Combined effect
SGOT	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.08	1.85	1.12	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.10	1.83	1.15	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.08	1.84	1.13	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.10	2.55	0.82	A
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.76	2.48	1.11	D
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.79	2.52	1.11	D
SGPT	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.92	1.37	1.40	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.92	1.37	1.40	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.92	1.43	1.34	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.92	1.37	1.40	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.92	1.60	1.20	D
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.92	1.54	1.24	D
ALP	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.11	2.06	1.02	D
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.38	1.74	1.37	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.87	1.90	0.98	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	2.11	1.82	1.16	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	1.72	1.86	0.93	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.95	1.84	1.06	D
LDH	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	3.31	2.19	1.51	S
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	3.39	2.23	1.52	S
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	3.86	2.32	1.67	S
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	3.95	2.45	1.61	S
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	4.37	2.69	1.63	S
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	4.48	2.81	1.59	S
GGT	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.90	1.77	1.64	S
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	3.10	2.00	1.55	S
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.96	2.13	1.39	D
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	3.17	2.30	1.38	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	3.22	2.47	1.31	D
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	3.44	2.40	1.44	D

Combined effects: CF; additive effects ($1 \leq CF \leq 1.5$): D; synergistic effect: ($CF \geq 1.5$) S; suppressive effect ($CF \leq 0.99$): A

including insulin signaling, glucose uptake, and glycogen storage in the hepatocytes. Thus, a disorder in glucose metabolism might increase its levels in the blood. These findings are consistent with those of Zhang et al. (Z. Zhang et al., 2021), who demonstrated that exposure to phthalates (PAEs) can induce disturbed biochemical homeostasis in fish and increase in oxidative stress following exposure. (Banaee et al., 2023a) observed that co-exposure to MPs and enrofloxacin (ENR) increased glucose levels and oxidative stress in *Oncorhynchus mykiss*, highlighting synergistic toxicity. (Lee et al., 2023) found that *Pseudobagrus fulvidraco* exposed to PE-MPs (>5000 mg L^{-1}) exhibited hyperglycemia and oxidative stress, indicating glucose mobilization under stress. A significant increase in glucose was reported in the plasma of rainbow trout exposed to MPs and *Yersinia ruckeri* infections (Banihashemi et al., 2022).

(Hu et al., 2016) showed that the liver, gills, and kidneys represent the primary metabolic pathways for phthalate esters. Thus, the ROS produced during the metabolism of xenobiotics may have destroyed of renal glomeruli (Multisanti et al., 2024). Findings from the present study indicated that exposure to 20 mg L^{-1} MPs resulted in elevated creatinine levels in the bloodstream of the goldfish. Notably, there was no statistically significant change in creatinine levels following exposure to 2.5 μL L^{-1} DPP combined with 40 mg L^{-1} MPs. However, the simultaneous presence of DPP and MPs increased creatinine levels. Furthermore, the findings indicated that DPP and MPs could significantly influence on the renal function and metabolism of fish, which may, in turn, affect the levels of creatinine. The present study showed that high concentrations or prolonged exposure to DPP and MPs may result in increased creatinine levels in fish, which could be indicative of potential kidney damage or impaired renal function (Banihashemi et al., 2022; Banaee et al., 2024b, 2024c).

Studies showed that changes in triglyceride levels could disrupt the metabolic processes in fish exposed to xenobiotics (Banaee et al., 2019a; Jenkins et al., 2019). Moreover, significant increase in triglycerides could negatively affect the reproductive health of male fish. Also, excess triglycerides can accumulate in vital organs such as the liver and disrupt their normal functioning (D. Zhang et al., 2024). Extra triglycerides might be stored in the fish's adipose tissue, which affects their buoyancy. High triglyceride levels in fish blood co-exposed to MPs and DPP might impair muscle function in fish (Buerger et al., 2022). Cholesterol plays an essential role in the biosynthesis of steroid and corticosteroid hormones, such as cortisol (Strauss III and FitzGerald, 2019). Results

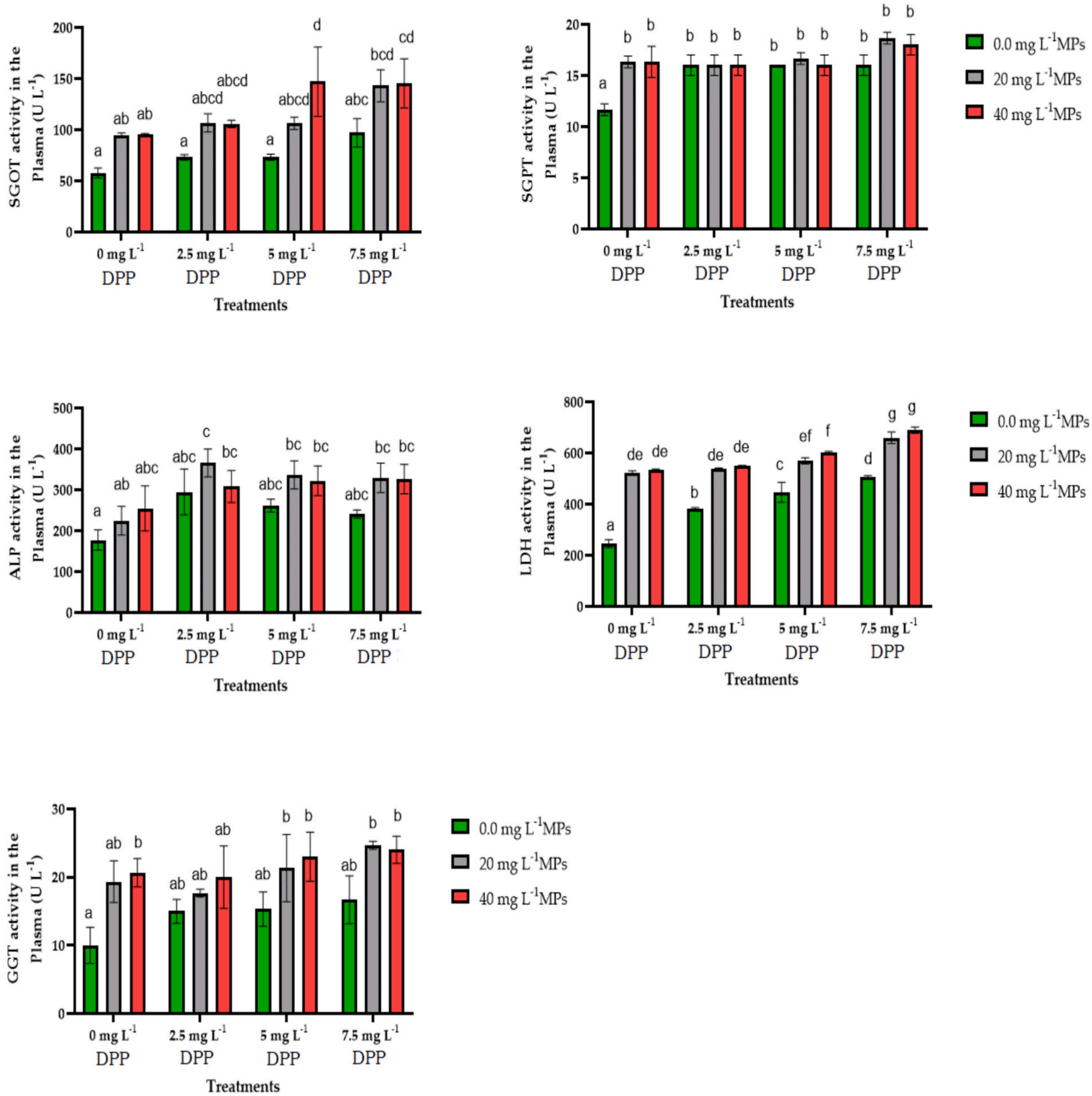

Fig. 3. Plasma enzyme activity levels (serum glutamic-oxaloacetic transaminase (SGOT), serum glutamic-pyruvic transaminase (SGPT), alkaline phosphatase (ALP), lactate dehydrogenase (LDH), and gamma-glutamyl transferase (GGT)) of male goldfish exposed to varying concentrations of microplastics (MPs: 0.0, 20 and 40 mg L^{-1}) and diphenyl phthalate (DPP: 0.0, 2.5, 5.0, and 7.5 μL L^{-1}), both individually and in combination, over 28 days. Different letters above the columns represent statistically significant differences between experimental groups as determined by Tukey's test ($p < 0.01$). Columns with the same letters indicate no significant differences.

showed that exposure to DPP and MPs could increase cholesterol levels due to disrupting the hormonal balance in fish, such as cortisol. Therefore, some studies showed that higher levels of cholesterol and triglycerides can suppress the fish's immune system, making them more susceptible to disease and infection (Bernardi et al., 2018). Furthermore, excess cholesterol in the blood may disrupt the optimal functioning of hepatocytes. Increased cholesterol levels in fish exposed to DPP and MPs could interfere with male fish reproduction by affecting the production and quality of sperm (Boopathi et al., 2023).

HDL is crucial for eliminating excess cholesterol from tissues through transport to the liver and excretion in the bile. Decreased HDL levels could inhibit the efficient removal of cholesterol, leading to its accumulation in organs and blood vessels of fish exposed to MPs and DPP (K. Zhang et al., 2024). HDL is involved in maintaining proper liver function (Banihashemi et al., 2022). Therefore, a significant reduction in HDL levels can compromise the liver's ability to metabolize lipids and detoxify xenobiotics efficiently. A decreased HDL indicated liver damage or dysfunction in fish exposed to MPs and DPP (Liao et al., 2023). The findings showed that fish exposed to DPP and MPs may experience an imbalance between LDL and HDL cholesterol, increasing the risk of cardiovascular issues (Zheng et al., 2024). Studies revealed that high LDL levels could be associated with increased inflammation and oxidative stress (Nnoruka et al., 2022). Inflammation can contribute to tissue damage, impair organ function, and disrupt normal physiological processes. LDL can interact with immune cells and disrupt their function (Hao et al., 2023), potentially leading to an impaired immune response in fish exposed to DPP and MPs. Elevated LDL levels in fish exposed to DPP and MPs may lead to the development of diseases impacting organs such as the liver and kidneys. Also, raised LDL could cause impaired organ function, compromised detoxification processes, and overall

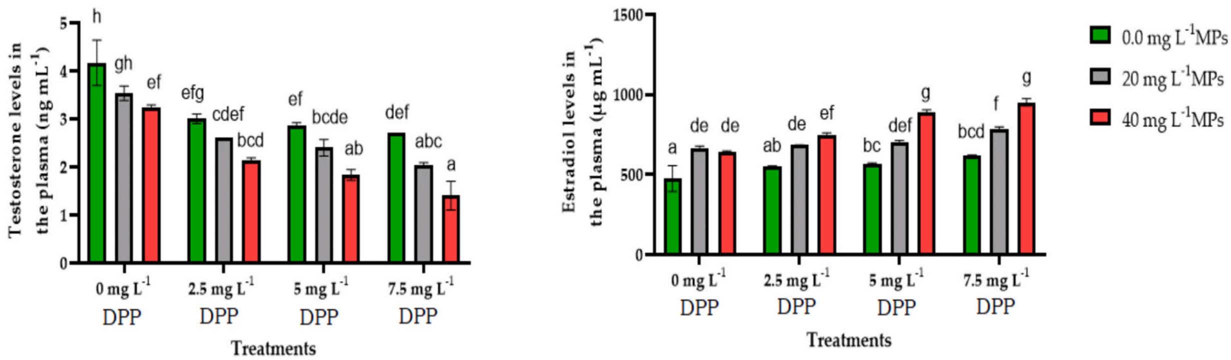

Fig. 4. Plasma hormone levels (testosterone and estradiol) in male goldfish exposed to varying concentrations of microplastics (MPs: 0.0, 20 and 40 mg L^{-1}) and diphenyl phthalate (DPP: 0.0, 2.5, 5.0, and 7.5 μL L^{-1}), both individually and in combination, over 28 days. Different letters above the columns represent statistically significant differences between experimental groups as determined by Tukey's test ($p < 0.01$). Columns with the same letters indicate no significant differences.

Table 4
Evaluation of synergism of microplastics (MPs) and diphenyl phthalate (DPP) concerning the plasma hormone levels (testosterone and estradiol) in male goldfish.

Parameters	Treatments	Predicated effect	Observed effect	Synergy ratio	Combined effect
Testosterone	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.31	1.40	1.66	S
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.55	1.44	1.08	D
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.40	1.48	1.62	S
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.61	1.57	1.03	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	2.61	1.66	1.58	S
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	1.75	1.88	0.93	A
Estradiol	2.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.28	0.65	0.44	A
	2.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.58	0.62	0.93	A
	5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.26	0.58	0.45	A
	5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.53	0.51	1.04	D
	7.5 μL L^{-1} DPP & 20 mg L^{-1} MPs	0.24	0.49	0.50	A
	7.5 μL L^{-1} DPP & 40 mg L^{-1} MPs	0.49	0.44	1.12	D

Combined effects: CF; additive effects ($1 \leq CF \leq 1.5$): D; synergistic effect: ($CF \geq 1.5$) S; suppressive effect ($CF \leq 0.99$): A

organ damage in the affected fish. Li et al. (Li et al., 2025) found that co-exposure to polystyrene microplastics (PS-MPs) and 17α-methyltestosterone (MT) in *Danio rerio* disrupted lipid metabolism. Lee et al. (Lee et al., 2023) demonstrated that polyethylene (PE) MPs (>5000 mg L^{-1}) caused bioaccumulation and elevated plasma cholesterol and triglycerides levels in *Pseudobagrus fulvidraco*. Moreover, Banaee et al. (Banaee et al., 2023a) observed that co-exposure to MPs and enrofloxacin in *Oncorhynchus mykiss* increased triglycerides and cholesterol levels. In contrast, Li et al. (Li et al., 2024) found that 2-ethyl hexyl diphenyl phosphate (EHDPP) could decrease cholesterol and triglyceride levels through changes in glycerolipid metabolism, steroid biosynthesis, and fatty acid biosynthesis pathways.

Forner-Piquer et al. (Forner-Piquer et al., 2018) showed that exposure to Di-isononyl phthalate and bisphenol A as endocrine disruptors could increase lipids and triglycerides and decrease the glycogen and phospholipids in the liver of gilthead sea bream, *Sparus aurata*. They also found that these plasticizers could change the expression of genes involved in the lipid metabolism in fish. The synergistic effects observed for cholesterol and creatinine levels at higher concentrations of DPP and MPs showed increased toxicity due to combined exposure. Additive effects were dominant for parameters such as triglycerides, LDL, and glucose, indicating that combined exposure exerted a cumulative effect without significant interaction. In contrast, suppressive effects for HDL and some glucose levels at higher concentrations may reflect the physiological constraints of the organism to maintain homeostasis under stress. The synergistic increase in cholesterol levels could indicate impaired lipid metabolism, possibly related to oxidative stress and liver dysfunction caused by the endocrine-disrupting properties of DPP and physical interference of MPs. Suppressive effects on HDL at higher exposure levels are of concern, as they may impair lipid transport and metabolism in fish. Similarly, the synergistic increase in creatinine indicated renal failure and highlighted the vulnerability of the kidney to combined toxins.

The results showed that exposure to DPP and MPs significantly reduced the total protein levels in the goldfish plasma. These findings indicated DPP and MPs had a potential for adverse effects on protein synthesis or stability. Sub-lethal exposure to DPP and MPs may hurt the liver and cause dysfunction, given the liver's crucial role in protein production. The reduction in total protein levels in the plasma of fish exposed to DPPs and MPs may be related to impaired protein synthesis in the liver, reduced intestinal amino acid absorption, and changes in cellular metabolism and amino acid transamination (Liu et al., 2024). Moreover, the interaction of DPPs and MPs with the cell membrane may significantly affect the process of amino acid transport into the cell cytoplasm or the secretion of proteins synthesized in the cell. Results showed that exposure to MPs and DPP could significantly affect albumin levels. Significant changes in albumin may be related to its physiological role in transporting phthalate and bisphenol in the blood (Lv et al., 2021; Vinod et al., 2024; Xu et al., 2024). The increase in albumin levels could be attributed to the fish's immune system reacting to the presence of these substances. Also, MPs or DPP might interfere with the metabolism or clearance of albumin in goldfish. They could impact the liver's ability to recycle or degrade albumin, leading to its accumulation in the plasma. Alternatively, they could affect the filtration of albumin in the kidneys, resulting in increased levels in the circulation. DPP and MPs could include disruptions to hormone regulation, nutrient uptake, or liver function, which can, in turn, affect albumin synthesis and secretion. The result suggested that a higher concentration of DPP may have a suppressive effect on globulin production in fish. Furthermore, exposure to MPs alone or combined with DPP reduced globulin levels. This result indicated that MPs alone and mixed with DPP could impair globulin production in fish. Globulin is another type of blood protein that plays a crucial role in the immune system as an antibody carrier and in fighting infections. Therefore, changes in globulin levels could have implications for the fish's immune response and overall health (Gholamhosseini et al., 2021). A significant disruption in protein metabolism was reported in zebrafish (*Danio rerio*) exposed to Polystyrene (PS) microplastics combined with 17α-methyltestosterone (MT) (Li et al., 2025). Lee et al. (Lee et al., 2023) found that exposure to polyethylene microplastics (PE-MPs) at high concentrations significantly decreased total protein levels in juvenile Korean bullhead (*Pseudobagrus fulvidraco*). The results showed that exposure to DPP and MPs significantly reduced immunoglobulin biosynthesis in fish. They also suggested that a higher DPP dosage could suppress fish immunoglobulin production. Immunoglobulins are essential antibodies produced by fish's immune systems to defend against infections and diseases. Therefore, changes in immunoglobulin levels can reduce fish's immune response to pathogens. Immunosuppression was reported in the fish exposed to phthalates (PAEs) (Z. Zhang et al., 2021; Y. Zhang et al., 2021).

Cell membrane damage due to an increased rate of lipid peroxidation may lead to the leakage of cytoplasmic and transmembrane enzymes into the blood (Banaee et al., 2023e, 2022). Increased SGOT and SGPT activities in the plasma of fish exposed to DPP and MPs could harm their health. These elevated SGOT and SGPT activities indicate liver damage and are often associated with toxic exposure. Therefore, fish with higher SGOT and SGPT levels may experience impaired liver function, compromised detoxification processes, and overall organ dysfunction. Increased ALP activities in the blood of fish exposed to DPP and MPs may indicate stress or tissue damage. ALP is involved in the digestion of dietary phosphates and absorbing essential nutrients. Increased ALP activity may disrupt these processes, leading to poor nutrient uptake and malnutrition (Banaee, 2020). LDH is present in various organs, and increased LDH activities in the blood may indicate organ-specific damage or dysfunction. LDH is involved in anaerobic pathways of energy production. Therefore, an increase in LDH activities indicated alterations in the metabolism of lactate and pyruvate, which could result in reduced energy production or inefficient use of energy resources within the fish's body exposed to DPP and MPs. Also, elevated LDH activities can be associated with oxidative stress. GGT is primarily found in high concentrations in the liver, and its elevation is often associated with liver injury or dysfunction. A significant increase in GGT activities may indicate that the fish's liver is experiencing adverse effects from exposure to DPP and MPs, such as inflammation or toxicity. Studies showed that GGT is involved in glutathione metabolism, an antioxidant molecule crucial in cellular defense against oxidative stress (Bachhawat and Yadav, 2018). A significant increase in GGT may indicate an increase in extracellular glutathione metabolism to enhance the cellular antioxidant defense system. Also, increased plasma GGT activities may be a physiological response to increase the rate of detoxification of xenobiotics (Banaee et al., 2024b, 2019b, 2024d). Li et al. (Li et al., 2025) demonstrated that PS-MPs and 17α-methyltestosterone (MT) caused liver damage and raised AST and ALT. Lee et al. (Lee et al., 2023) reported that PE-MPs increased AST, ALT, and ALP in a concentration-dependent manner, reflecting liver damage. Li et al. (Li et al., 2023) found similar effects of PS-nanoplastics on AST and ALT in zebrafish. Banaee et al. (Banaee et al., 2023a) showed that enrofloxacin and MPs co-exposure elevated AST, ALT, and ALP, enhancing pollutant toxicity. Banihashemi et al. (Banihashemi et al., 2022) observed that MPs combined with *Yersinia ruckeri* increased AST, ALT, ALP, and LDH, indicating liver damage and oxidative stress. Liu et al. (Liu et al., 2024) found that a significant increase in AST (SGOT) and ALT (SGPT) in the serum and liver could be related to hep6atocyte vacuolation and nuclear pyknosis of zebrafish after exposure to 2-ethyl hexyl diphenyl phosphate. Yang et al. (Yang et al., 2018) found that exposure to di(2-ethylhexyl) phthalate could increase the risk of oxidative damage, neurotoxicity, and apoptosis in fish. These findings highlight that MPs exacerbate the toxicity of other pollutants, leading to elevated liver enzyme activities and oxidative stress in aquatic organisms. The effects of DPP and MPs on SGOT and SGPT activities were additive, LDH and GGT, while their impact on ALP activity was concentration-dependent.

The effect of DPP on estradiol levels in the blood of male fish can be significant and concerning. Increasing estradiol in male fish exposed to DPP and MPs may suggest potential hormonal disruption or effects on the endocrine system (Forner-Piquer et al., 2018). Elevated estradiol levels in male fish could lead to reproductive abnormalities or disruption of normal sexual characteristics (Scholz and Klüver, 2009). DPP can act as an estrogen mimic or disruptor, binding to estrogen receptors and triggering estrogenic effects. These findings can lead to increased estradiol production, a form of estrogen, in male fish. Elevated estradiol levels in male fish may result in feminization effects, such as developing female reproductive characteristics or altered reproductive behaviours.

Results showed that exposure to DPP and MPs could change the normal testosterone levels in male fish by disrupting hormone biosynthesis (Toft et al., 2003). Reduced testosterone levels in male fish exposed to DPP and MPs can lead to reduced sperm production, altered reproductive behaviours, and impaired secondary sexual characteristics. The study has shown complex interactions between MPs and DPP on plasma hormone levels in male goldfish. Synergistic effects on testosterone were observed at lower concentrations of MPs and DPP, suggesting enhanced toxicity when combined. However, higher concentrations resulted in suppressive effects, possibly due to toxic overload or interference with hormone production. Estradiol levels showed suppressive effects at lower concentrations and additive effects at higher concentrations, indicating potential steroidogenesis and receptor binding disruptions.

5. Conclusion

The study findings indicated that the concurrent presence of DPP and MPs in fish environments could exert a synergistic effect on glucose levels, potentially disrupting processes involved in glucose metabolism. Moreover, exposure to DPP and MPs has been observed to affect creatinine levels, which may indicate the presence of kidney damage or impaired renal function. Furthermore, the research indicates that DPP and MPs may impair protein synthesis, which could potentially lead to liver damage or dysfunction. Furthermore, DPP and MPs may induce a stress response, inflammation, and alterations in albumin, globulin, immunoglobulin, triglyceride, and cholesterol levels, which could affect the fish's immune response, energy regulation, reproductive health, and organ function. An elevation in the activities of SGOT, SGPT, ALP, LDH, and GGT in plasma may be indicative of liver damage, impaired detoxification, tissue damage, alterations in metabolism, or oxidative stress. In conclusion, exposure to DPP and MPs could have far-reaching consequences for the health and well-being of fish. Moreover, the results highlighted the disruptive impact of DPP and MPs on testosterone and estradiol levels in male fish, underscoring the necessity for continued research and monitoring to comprehend and mitigate the potential ecological consequences.

CRediT authorship contribution statement

Masoumeh Faramazinia: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Gholam Reza Sabzghabaei:** Methodology, Investigation, Formal analysis, Data curation. **Cristiana Roberta Multisanti:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Mahdi Banaee:** Writing – original draft, Visualization, Supervision, Methodology, Conceptualization. **Giuseppe Piccione:** Writing – review & editing, Visualization, Validation. **Abha Trivedi:** Writing – review & editing, Software, Resources, Formal analysis, Data curation, Conceptualization. **Caterina Faggio:** Writing – review & editing, Supervision, Project administration.

Ethical approval

The Ethics Committee of the Behbahan Khatam Alanbia University of

Technology in Iran approved the experimental procedure and methodology of this study (Approval ID: 6-1-112667/13-03-02).

Funding

This research received no external funding.

Declaration of competing interest

The authors declare they have no conflict of interest.

Data availability

Data will be made available on request.

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